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A survey on Blockchain based pharmaceutical supply-chain management and drug distribution : The case of Morocco

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***Abstract*— The pharmaceutical supply chain management is facing numerous challenges that had led in the past and still today to issues in the distribution of drugs and their availability in the market. When interviewed, stakeholders affirm the issue does not involve the scarcity or lack of production, but indeed inventory allocation. The complexity of the drug supply chain fabric and the critical nature of these synergies' request for a holistic solution that should be able to assure both the validity of the data and the durability of the records. Blockchain appeals to be the right solution for this problem. In this paper, we will be conducting a quantitative survey on the use of blockchain technology in the proper management of pharmaceutical supply chain. We will be pointing out the main current challenges such approaches face, and finally, we will be projecting the on the feasibility and underlining the applicability to the case of a pharmaceutical supply chain in Morocco.**

Keywords—Blockchain, Pharmacy, Supply-chain, healthcare, security.

I. Introduction

The pharmaceutical supply chain is one of the most critical processes in the field of healthcare services, yet undoubtedly one that receives the least attention in the current research stream for improved and smart health undertakings. Distribution management malfunction had taken a toll on the quality of services provided to the patients as it had been reported to be the cause of major drug unavailability events rather than what had been blamed on an undersupply of medication. As the reoccurring of these events impact the right of access to healthcare to very critical patients, it needs immediate remediation. To answer this need, blockchain had been helping, through the last decade, to evolve critical processes from centralized and complex, to decentralized and versatile. It grants the elimination of Single Points of Failure (SPoFs) by managing a distributed database of records between many owners and allowing as well for more transparency and better traceability, being the key missing attributes to allow clinical pharmacies to predict, and thus overcome episodes of drug unavailability, and that could allow clinical pharmacists to track and better manage the inventory of medications, especially in cases of incurable diseases that needs constant availability of the medication on the shelves.

A. Motivation

This study had been motivated by recurring events of mediation hiatus from certain markets in Morocco in the recent years. The last most impaction occurrence of this nature had been observed amid the raise of reported cases during the first and second wave of the Covid-19 pandemic, when the shortage of supplements and anti-inflammatory medications resulted in an alarming crisis and panic among the population. Another occurrence of this phenomenon had been commented on more recently, involving a reported undersupply of Levothyroxine, the medicinal compound replacing the missing thyroid hormone thyroxine for patients. Distribution management malfunction had been reported to be the cause of this event rather than an undersupply of the medication. This observation led us to conduct research about secure solutions that can help resolve this issue, combining reparable tracking ability, auditability, and temper resistance, research trends are positioning blockchain as one of the most promising solutions to this matter.

B. Our Contribution

Our survey will focus on providing a comprehensive overview on the use of blockchain in the drug distribution system overall, focusing on a later part on the reduced territory of the Moroccan kingdom, we will aim at a first place to identify the state-of-the-art studies on the matter, we provide an analytic, quantitative review on the literature, and we will be adducing a critical guideline to help create a decentralized, up to date, verifiable, and temper proof database helping pharmacists, clinical pharmacy managers and hospitals to record, track, and thus supervise more efficiently the drug distribution in the territory.

C. Related Works

Amid searching for literature around the subject, we have noticed a severe lack in literature reviews regarding blockchain applications to pharmaceutical supply chain management. We have come across [1] where authors have conducted a similar quantitative study, that had been more focused on the statistical aspect of the research and focused more on blockchain than in the medical supply chain, where Researchers addressed the importance of blockchain into creating a new era for electronically processed medical data, with its assets targeting safety and traceability, genuine records and none repudiation ensuring the authenticity of the data and its provenance. There had been several studies and reviews that had been conducted on the application of blockchain to the supply chain management counting [2] that exhibits most of the solution portrayed in the literature are meant to product tracking deterrence of fraud, while pointing information asymmetry and quality of data injected into the blockchain as main blockchain as a main concern that I yet to be addressed by the literature that I overall still very reluctant to embrace this technology, and [3], but shall be extended to the specific case of medical supply chain. Overall, many other

instances of the literature agree that medicine's supply chain in Morocco is in need of a rerouting and modeling of conception of its core elements that should be anchored in the vision of digitalization that the country is leveraging, by designing new models, [4] and adjusting transportation strategies, targeting components that may improve the quality of the products and the efficiency of the drug dispensing and transport [5].

Another issue that will be discussed in this paper is the drug counterfeit, this mayhem targeting the pharmaceutical industry, costing millions of dollars losses and that can lead to life threatening events in consumers was addressed by [6] that proceeded to establish a holistic overview of the existing schemes that addressed the matter, as well as pinpointing the pitfalls that can be alleviated upon the usage of blockchain, which was complemented by other studies where the focus was on the aspect of assessment of the efficiency of this technology [7] as a promising solution of many troubles complex chains of value as that of pharmaceutical supply chain faces, whereas [8] investigated the novel governance paradigm that is forming around IoT and blockchain to improve pharmaceutical blockchain as a part of a more efficient drug governance effort, where they stated the potential this combination of technologies can provide for a more reliable and efficient drug distribution system.

In the territorial context of our research, we came across [9] where authors surveyed the state of function of the Moroccan supply chain of medication, the difficulties and malfunctions that affects the system that authors reviews included optimizing the flows and stocks management, they also stated the improvement of information systems that carries the data between the facilities, as the main target of improvement to create optimal solutions.

D. Paper organization

Our survey will be organized as follows: In the first section, we will be introducing our motivations and contribution with this study, as well as mentioning the related works. We will then exhibit the correlation between blockchain and medical supply chain management. After which we proceed to a review of the found literature, and then we will discuss the different cited and reviewed challenges, and we will conclude with our perspectives for future works.

II. BLOCKCHAIN APPLIED TO DRUG SUPPLY CHAIN MANAGEMENT

Openness had been a key pillar of governance in the smart city paradigm the collective effort of technology is heading towards. It is as well a main component of smart health, as a much crucial part of the creation of this smart society and urban change, as imagined by Boyd Cohen is his smart city wheel [10]. Nevertheless, smart healthcare had been at the center of the research subject that aim to improve the access of citizens to better and thriving healthcare services. Blockchain has its fair share of the discussion, where it had been discussed as a main component of smart healthcare systems that merges with Internet of Things, addressing many of its shortcomings [11].

Blockchain technology has revolutionized transactional activities and had been recognized as a game-changer in the financial industry since its invention in 2008, when it was later introduced as the backbone of the cryptocurrency.

However, its potential extends far beyond the financial sector, as it had been studied to provide solutions to a variety of problems in different industries, such as healthcare, supply chain management, and voting systems. One of the key features of blockchain technology is its decentralized nature, that prevents single entities from exercising any total control over the network, by providing every participant of the ledger with a copy of up-to-date record, thus preventing malicious participants from taking over the system. This feature provides a level of transparency and security that is unmatched by traditional centralized systems that were very susceptible to single point of failures and hard to audit.

Another important aspect of blockchain that makes it more secure and private is the use of cryptography, where transactions are encrypted and verified by multiple participants. In addition to its security and transparency, blockchain has also inspired a new paradigm of ownership and control, that is distributed among the stakeholders, making it much more participative.

Transparency and accountability are two of the strongest arguments of blockchain, where instead of limiting accountability to the actions of a single entity. It is again distributed among the nodes in the network.

A. Tranformative imapcts of Blockchain in medical supply chain

The pharmaceutical industry development and services overall rely on loads of data from different types and sources that serve many purposes that offload from the drug development, testing, until the distribution to pharmacies and medical facilities, that go through a very complex tissue of structures and environments. Unfortunately, situations of crisis like the Covid-19 pandemic have exposed the disruptive nature of poor distribution channel management of medication, and the magnitude of the negative impact compromised supply chain management can have of the welfare of the citizens. The pharmaceutical supply chain management is a complex fabric of interlocked processes that involves multiple stakeholders, including manufacturers, wholesalers, retailers, and healthcare providers. However, the current system is plagued by inefficiencies that lead to seemingly unresolvable challenges, such as, inadequate tracking, poor visibility, counterfeit drugs propagation...etc. Hence, a more efficient approach to the existing system should be able to provide:

- An immutable, transparent, and secure record
- A better tracking system Ability to audit.

Blockchain technology has the potential to address these challenges and enhance the medication supply chain management process. The distributed database nature of the ledger allows all parties to be involved with complete transparency in the supply chain management by creating a room for consensus upon updating the ledger, establishing, and following the same protocol standard. One of the key benefits of using blockchain in this case is the ability to track drugs from the point of manufacture to the point of consumption. Each transaction along the supply chain is recorded on the blockchain, creating a tamper-proof and transparent record of the drug's journey. This enables all parties a real-time visibility into the chain, reducing the risk of undersupply and shortage as well as counterfeit medications entering the system. This extensive use of data can extend to patient consumption data, and agents management of medical waste, preventing hazardous

propagation of unused medication later in the lifecycle with the means of a better drug traceability system, based on transfer of ownership of the product Additionally, the use of smart contracts and digital identities allows the automation of drug tracking, making it more efficient and reducing the need for manual intervention, as well as the benefit improved trust among stakeholders, reducing the risk of fraud and corruption, especially with the use of cryptography and digital signature, providing a secure and private way to exchange information.

Furthermore, blockchain can be used to improve the supply chain's responsiveness to changes in demand. Realtime visibility into the supply chain might help parties a quicker identification of potential issues and allows furtive adjustments to maintain a steady supply of medication. This can help reduce occurring episodes of stockouts and ensure that patients, especially in critical condition or suffering chronic diseases, have proper and supplemented access to the medication they rely on. This led to the creation of many models that target the integrity of the medication supply chain, such as proposed by [12] using and accessing the properties of private blockchains like Hyperledger Fabric.

Another groundbreaking usage for the technology of blockchain might be product lifecycle management, and especially the end-of life management of hazardous materials [13] that is directly applicable to the byproducts of Pharmaceutical industry.

Upon a study conducted by the World Health Organization, alarming numbers suggest that 1 in 10 drugs available in the developing countries markets of medical product is a fake medicine, putting a huge proportion of already fragile population at risk if no efficient recall system is established. Mitigating the loopholes that led to these situation, Blockchain had been proposed in several instances, such as by the authors in [14] to preserve both the suppliers, manufactures from losses, and the patients from the severe consequences of drug misusage, that resorting to using Elliptic Curve Digital Signature to further enhance the security of their schemes, likewise, the issue of counterfeiting led the intellectual efforts into considering the merging of Blockchain with other technologies such as proposed by [15] following the drug delivery process, combined with machine learning, aiming to help costumers chose the best suited medication.

III. REVIEW OF THE LITTERATURE

After using the keywords to our research, we had been faced with a shortage of literature reviews and materials which substance addressed the use of blockchain as a new paradigm to the medical supply chain management in the pharmaceutical or drug dispensing realm. Yet, any application of Blockchain to supply chain management did not reach maturity and still encountering some challenges.

Author of [16] mentioned that some traceability solutions are stuck in the conceptual stage or lack specific and comprehensive effectiveness evaluation. Very few studies build the system from the bottom level up. Most of which are based on the mainstream blockchain platform. The other gives however the example of the Chinese Tmall Global that uses blockchain and product labels with QR codes to track food and Walmart announcing the result of its project on using blockchain technology to track and monitor mangoes in the United States and pork in China, implying as well that blockchain-based supply chain traceability solutions are preferably used to trace high-value products, such as diamonds and the above-mentioned high-end beef. And since complex supply chains have relatively large amounts of data, to reduce the load of blockchain and alleviate data explosion, some scholars proposed the mechanism of multiple storages. Regarding the pharmaceutical supply chain, authors in [1] mentioned the same constraint as per which drug supply management are still facing logistical challenges, especially faced with the uncertainty that the field faced during the outbreak, and where blockchain could be of a considerable potential in displaying real time information accessing the volume and location of medication and supplies as well as maintaining record history for administration and purchasing. [2] reviewed other articles that used Ethereum and Hyperledger Fabric as framework to their supply chain management systems, and pointed out the double chain system, that consists of a transaction chain for storing the transaction data and a user information chain for storing the enterprise information, inferring that the split-chain storage of the two data types improves the system's efficiency. In [17] authors had stated that due to the absence of a standard protocol to carry out the distribution and commercialization of medicines, they realized a proof of concept carried out to an application that maintains a registry of batches of vaccines necessary for COVID-19. Smart contracts had also been specifically mentioned in the literature by [17]. As a recommended mechanism to overcome theft, loss, and fraud problems as smart contracts can incur different costs and risks for health-care applications. The authors empirical analysis shows that smart contracts play a significant role in creating value for customers, retailers, and manufacturers, mentioning that despite the positive effects of adopting smart contracts on blockchain platforms in the pharmaceutical supply chain, smart contracts' potential to combat counterfeit drugs has not been fully exploited. Author mentioned as well MultiChain as the simplest blockchain platform to implement. Authors in [18] implemented a drug data storage system using blockchain. system uses Solidity programming language on the Remix platform as a proposed combination of drug data storage with access rules and encryption of stored data. In another review written by [3] Authors found that most of the solutions aim at the tracking and tracing of products or fraud prevention. Challenges include information asymmetry and unknown provenance. As well, quality of data input into the blockchain systems is not adequately addressed by current literature. And overall, the proposed solutions are in the metrics testing phase, and many have not implemented the proposed systems with firms. Thus, according to [16] To materialize the advantages of blockchains, it is not sufficient simply to implement the technology, since the technology only represents the infrastructure, a shared and functional information system would require set of rules to determine the parties that will be in charge of developing the system, those who have rights, the rights that they have, and how to integrate new members.

Authors in [19] stated that when one of the most powerful assts of Blockchain is the immutability, tamper resistance can be intolerant to human errors, that are undetachable from any human driven process. Decentralization seems also to create latency, and blockchains are very resource hungry processes, however, the strong link that had been found between information sharing and performance might be a benefit to the technology itself.

Finally, authors in [20] insists that, as blockchains cannot exist alone, and that it still needs infrastructure, as data bases, it will not be a factor that shrinks the number of IT interfaces. Consequently, when risk mitigation comes with its own cost, they see no added value to tracing and tracking, unless data immutability is mandated.

IV. Challenges and discussion

A concrete example of the obstacles that appeal to use blockchain as a solution for the medication supply chain management had been mentioned by [21] in a field study conducted in Morocco, according to which the major problems that face public hospitals during the acquisition and distribution of the medications are the following :

- Lack of management
- Lack of traceability
- Lack of supply planning
- No mechanism for measuring the supplier's KPIs
- Usage of paper compliance checks
- Management is manually done via stock records.

Author of [17] affirms that using blockchain can represent a huge milestone to the development of these situations, as paperless mechanism using blockchain provides an immutable and decentralized anti-fraud system. Smart contracts are being discussed as a system that uses blockchain technology. Hence, smart contracts can grant user access to track counterfeit returns to manufacturers and suppliers using RFID tags. Patients can also access real-time data to advance clinical trials.

However [22] affirm that the lack of suppliers with experience and the clash of interests between the different stakeholders could be other factors that prevent the adoption of these solutions as supply chain participants would generally prefer to keep the majority of their data confidential. There are several other recurring examples of challenges that affects the ability of using blockchain in pharma, and more specifically in the drug supply chain management field. But most of the studies agree that the most prominent sones might be the high cost of deployment, related to lack of expertise of the agents in the field. Regulations, either on the side of patient's data or around industry standards are major drawbacks to the implementation of such processes as well, and the problem of scalability share the issue of the cost of deployment and storage. However, foremost, the unwillingness of the parties to share their sensitive data can also be counted as a challenge to any measure of change.

However, it is a certitude that different stakeholders of the medical realm cannot fathom a future for technology enhanced supply chain without techniques such as blockchain, but are still counting on the development of targeted, tailored solutions rather than adapting existing solution. The main drawbacks that are in another way a

chance for in-house or tailored blockchains is the compliance with the varying or shallow indoor policies and laws surrounding processes and information disclosure. using blockchain. Drug supply chain involving a market where the concurrence is very tight, leads to the unwillingness of the different stakeholders to share data or collaborate with other concurrent on same platforms.

The authors in [23] went in this direction, by suggesting the use of Gcoin in the supply chain data flow, mentioning information delays and infrastructure breakdown, and lack of transparency as key points that can be mitigated and overcome as well as reducing the cost due to the improvement of the quality of data exchanged. As we mentioned earlier, inhouse solutions can be somewhat of a solution to these problems, but this includes building ledgers form the ground up, which again introduces its own sets of challenges including difficulty to program and maintain, the necessity of testing, that could take years of development, and that can be risky for a critical sector as healthcare where experimenting with sensitive data can be threatening. On that note, several technical aspects can be a deal breaker for this technology, including its inflexibility to errors, the difficulty to overcome or change ancient smart contract or chain code terms, and the still sensitivity to data forgery attacks that exploits the trust these schemes have in their authentication and submit data that is fallacious to the ledger, which is hard to target and to detect [24].

Another issue being the constraint new technologies bring to the healthcare professionals, that had and is still discussed among the main reasons that refrain the adoption of any new technology into healthcare processes.

V. Conclusion and perspectives

We aimed through this study to conduct quantitative research on the on the use of the blockchain technology in the management of pharmaceutical supply chain. We intended to first explore the state of the art regarding approaches using these technologies, then addresses the challenges similar approaches faced and are still facing to be properly deployed and exploited.

Blockchain withholds a huge potential to change the way we perceive supply chain management, to the best. By inducing better drug distribution, we can contribute to the holistic patient data management effort intended for a futuristic design of Smart Healthcare, and the evolution of research in the medical field by better intellectual property enforcement, and thanks to the reduced time to market of healthcare products the usage of smart contract might generate. Blockchain can undeniably help keep better track and provide more statistical information on side effects and help drive automated requests for more drug product in demand and ensure a better respect of the follow up protocols, and the supply chain can be made more transparent as controlled by all participants, shining the light of the constraints that are faced by the example of the Moroccan drug supply chain management that could be overcome with the adoption of this technology, as well as some of the challenges that constitutes its main drawbacks.

We will then extend this study, if appliable, to a feasibility study, involving the order of law regarding the matter of blockchain usage in medical facilities and pharmaceutical supply chain in Morocco, especially in the matter of end-oflife drugs and supplies, and address the challenges around the subject. We will then proceed to theorize the applicability of these solution in the drug supply chain in Morocco, as opposing to the law and regulations in this matter, and if applicable, analyze the factors holding back the integration of these solutions.

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